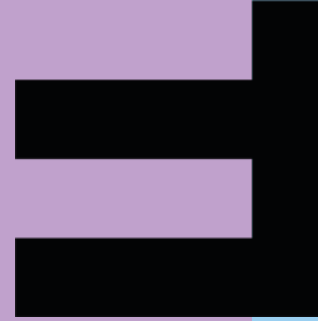




074703043202 - Energy transmission and distribution

Ruben Lliuyacc Blas
11.05.2026

About the lecturer



Ruben Lliuyacc Blas

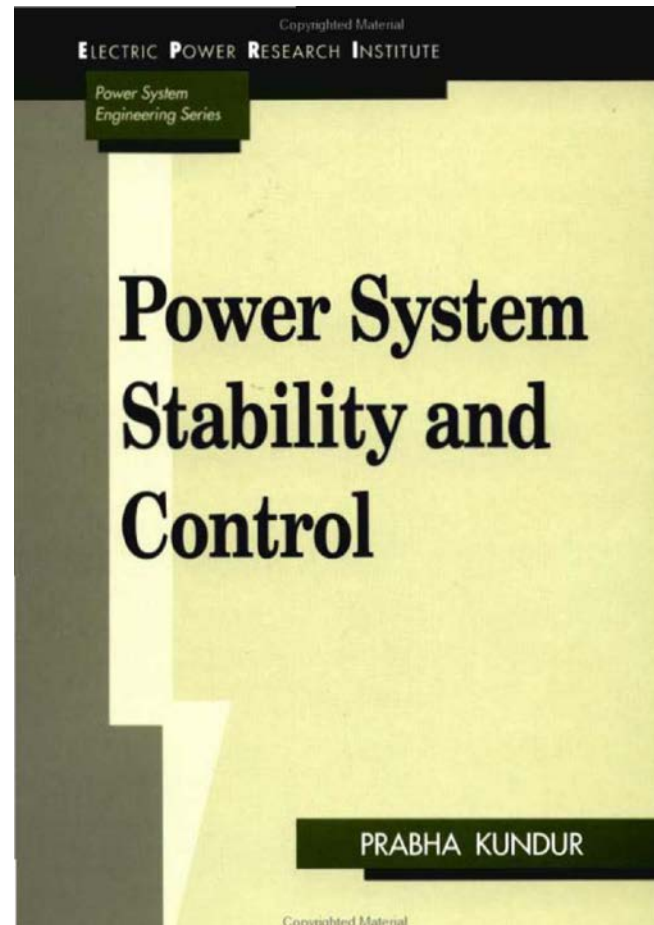
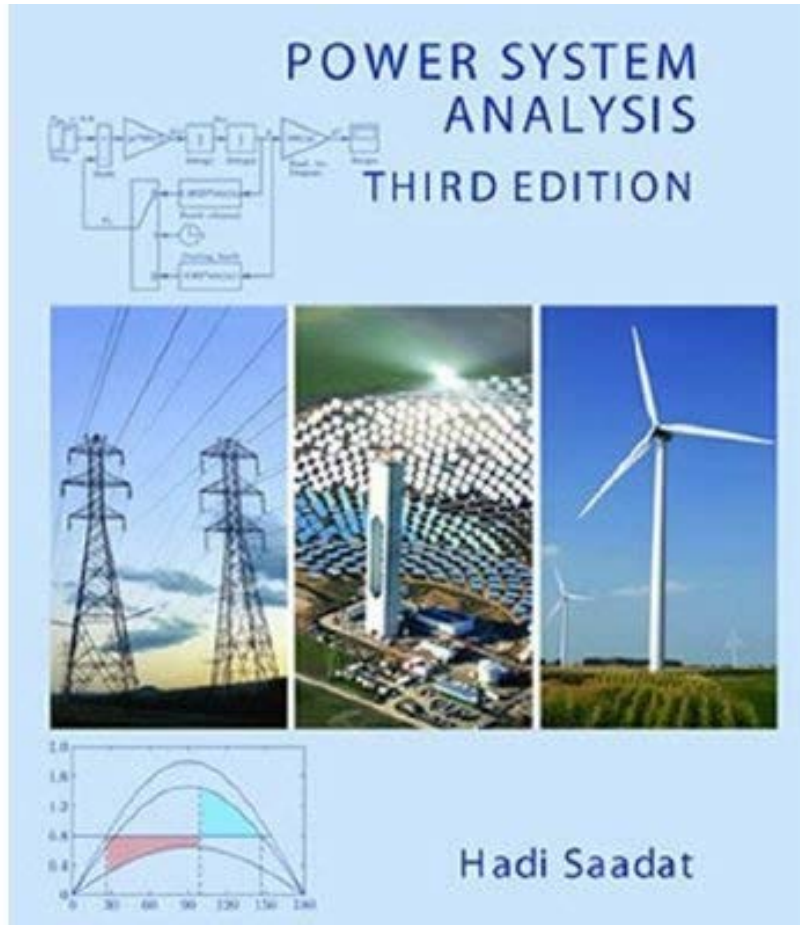
Postdoc, Energy Research Centre, FHV

<https://www.fhv.at/en/research/energy>

llru@fhv.at



Recommended books



... and also **AI**, but be **careful!**

Prerequisites

- **Basic of electric circuit** (sinusoidal voltage, rms values, phasor representation, solving basic electric circuits (resistance, inductance, capacitance)).
- **Concepts of three-phase systems** (line-to-line voltage, phase-voltage, line current, phase current, single-phase and three-phase transformers).
- **Basic of electric machines** (rotating and static, calculation of active power, reactive power).



Lecture (13:10 – 15:35)

- **13:10 – 13:50:** Lecture, part I
- **13:50 – 14:00:** Break
- **14:00 – 14:40:** Lecture, part II
- **14:40 – 14:50:** Break
- **14:50 – 14:35:** Lecture, part III



Lecture objectives

- Calculate **by hand** and using **computational tools** the steady state of simple and complex **power systems**.
- Understand the **inputs, outputs and parameter involved** to determine the steady state of power systems.



Content

Part I

Power system

- ☐ Components
- ☐ Three-phase system
- ☐ Per unit system
- ☐ Calculations

Part II

Power flow

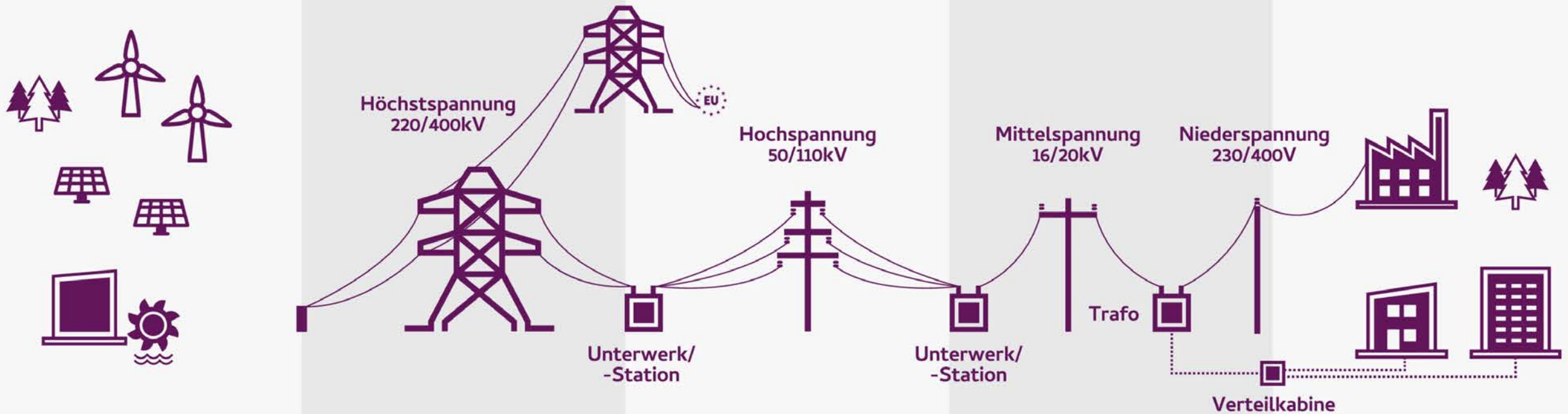
- ☐ Admittance matrix
- ☐ Power flow equations
- ☐ Jacobian calculation

Part III

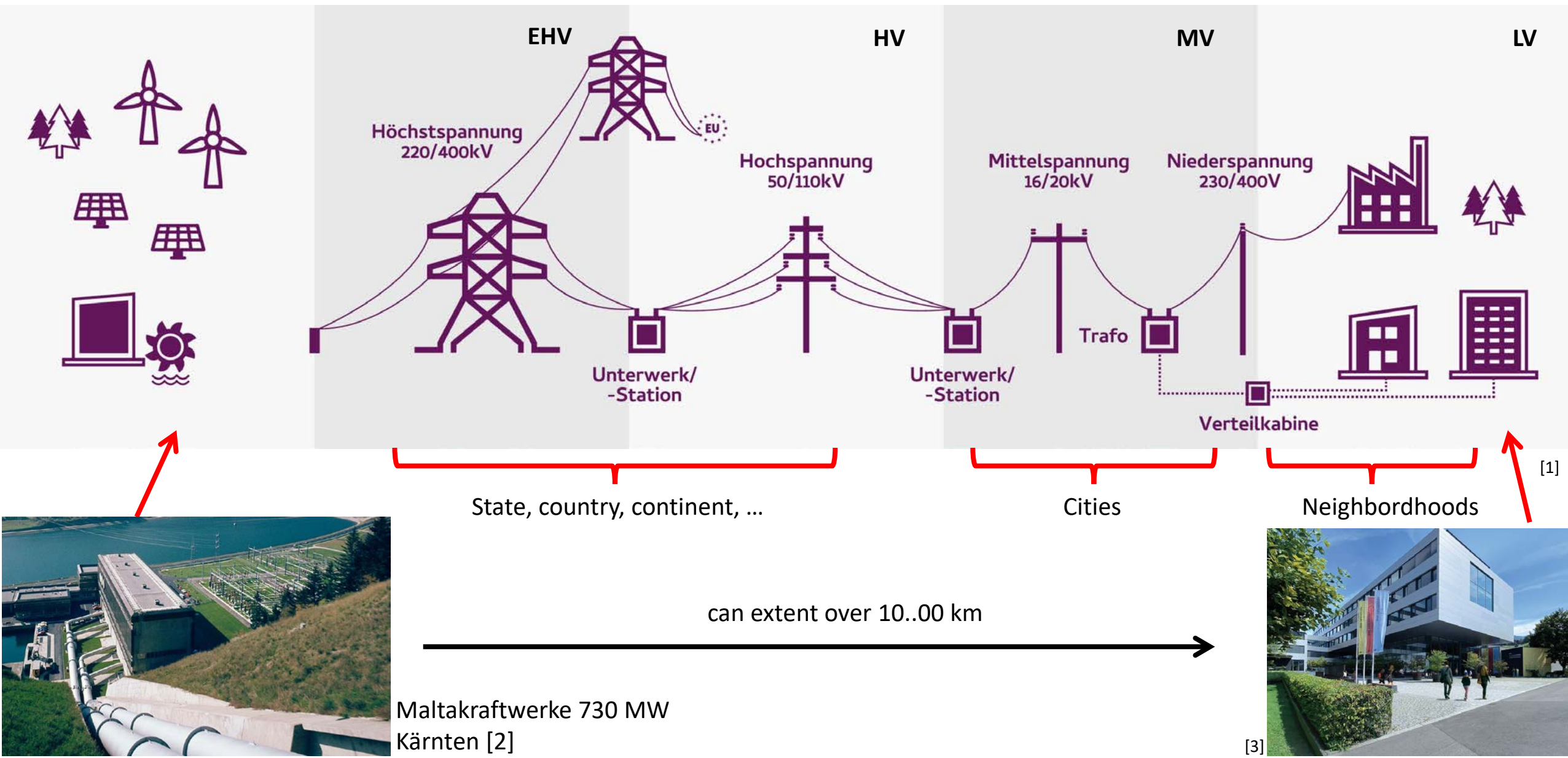
Bonus: Optimization in power system

- ☐ Unit commitment
- ☐ Linear optimization in power systems

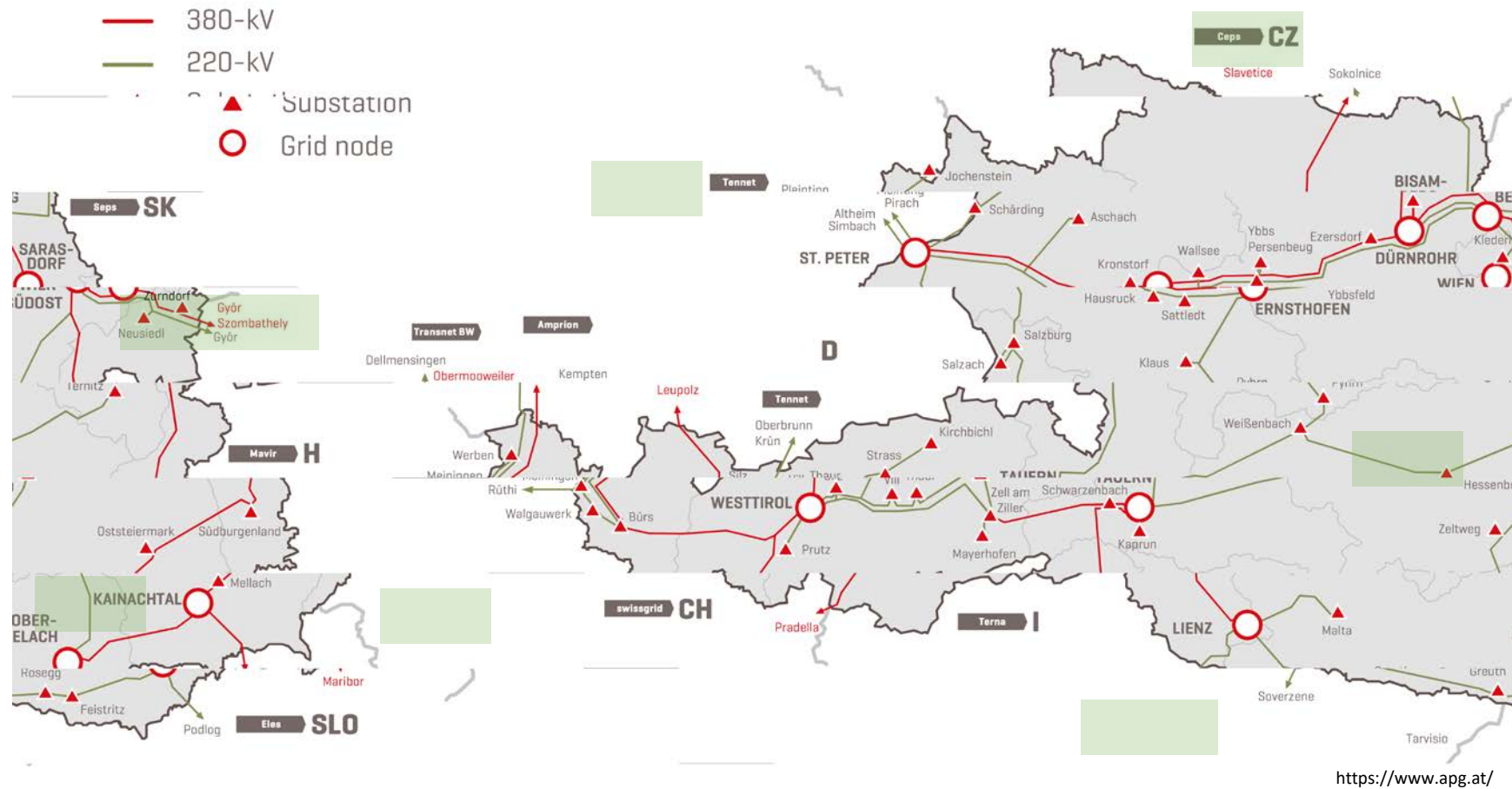
Part I: Power systems



Power system overview

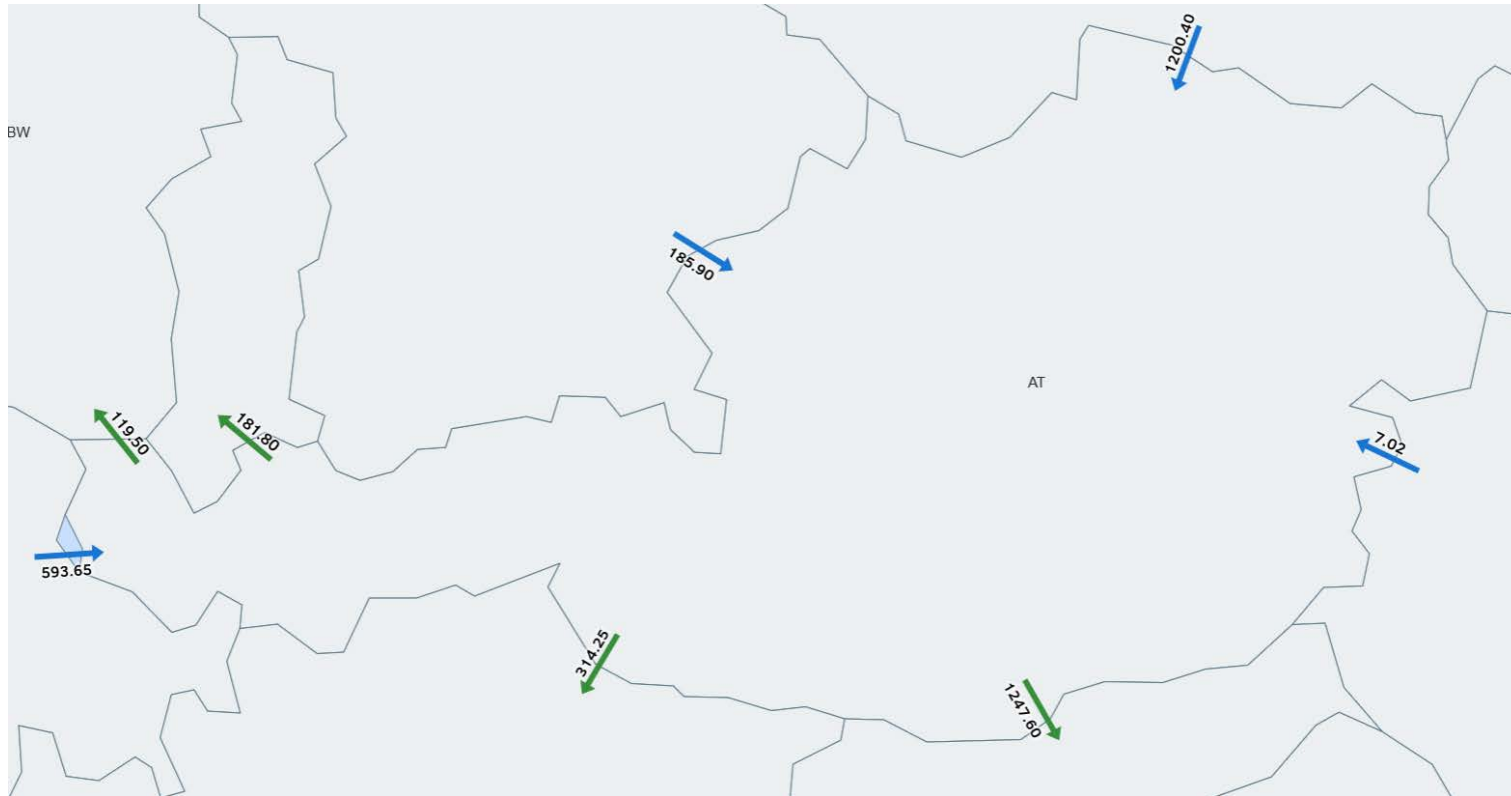


Power system



Power system

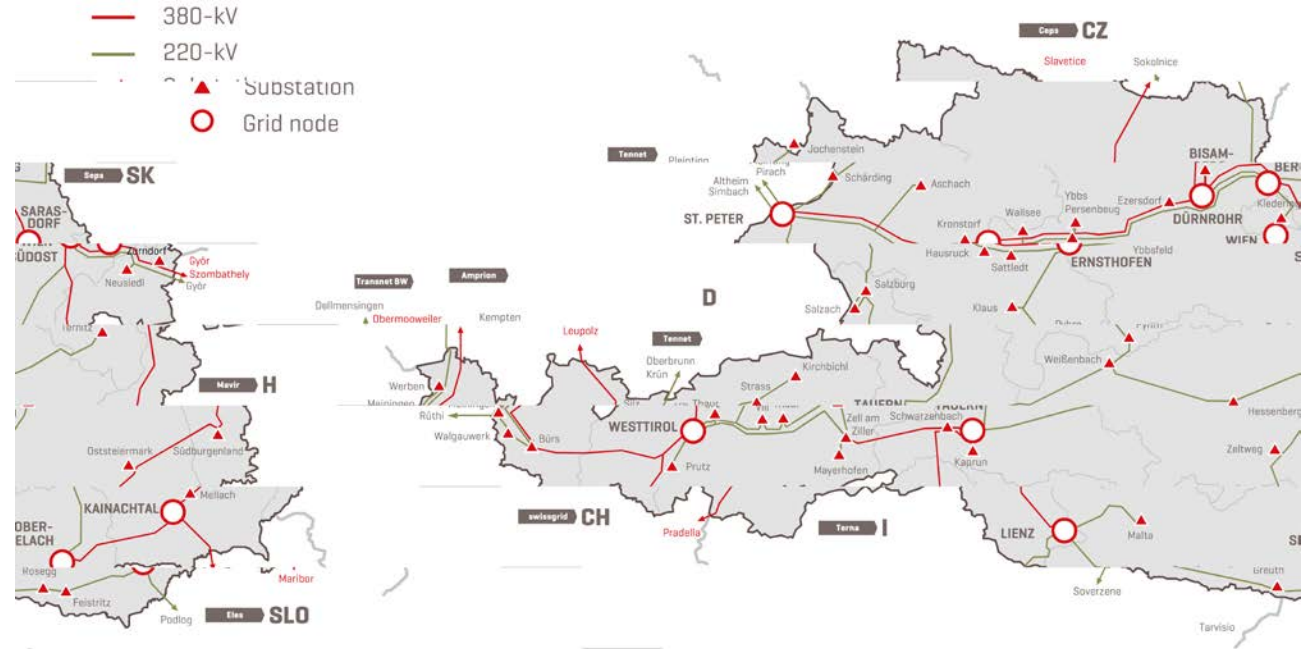
■ Export/import in MW: 10.05.2026



<https://transparency.entsoe.eu/>

Main components

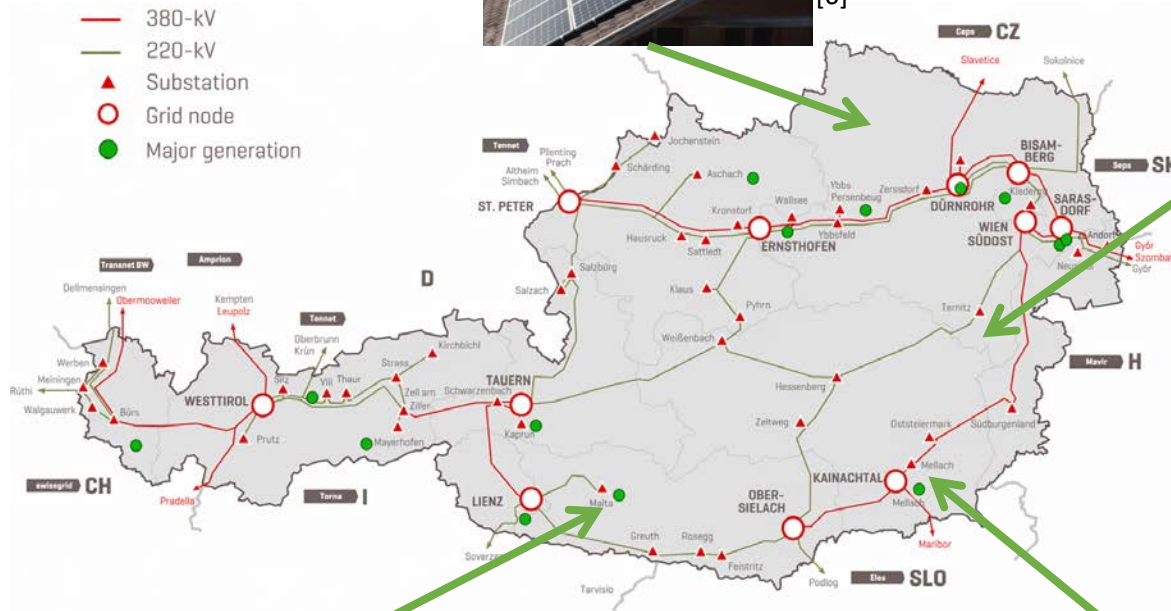
- ❑ Generation
- ❑ Network (EHV, HV, MV, LV)
- ❑ Transformers
- ❑ Demand



Generation



[6]



ChatGPT



Fürstkogel 3.45 MW, Steiermark [5]

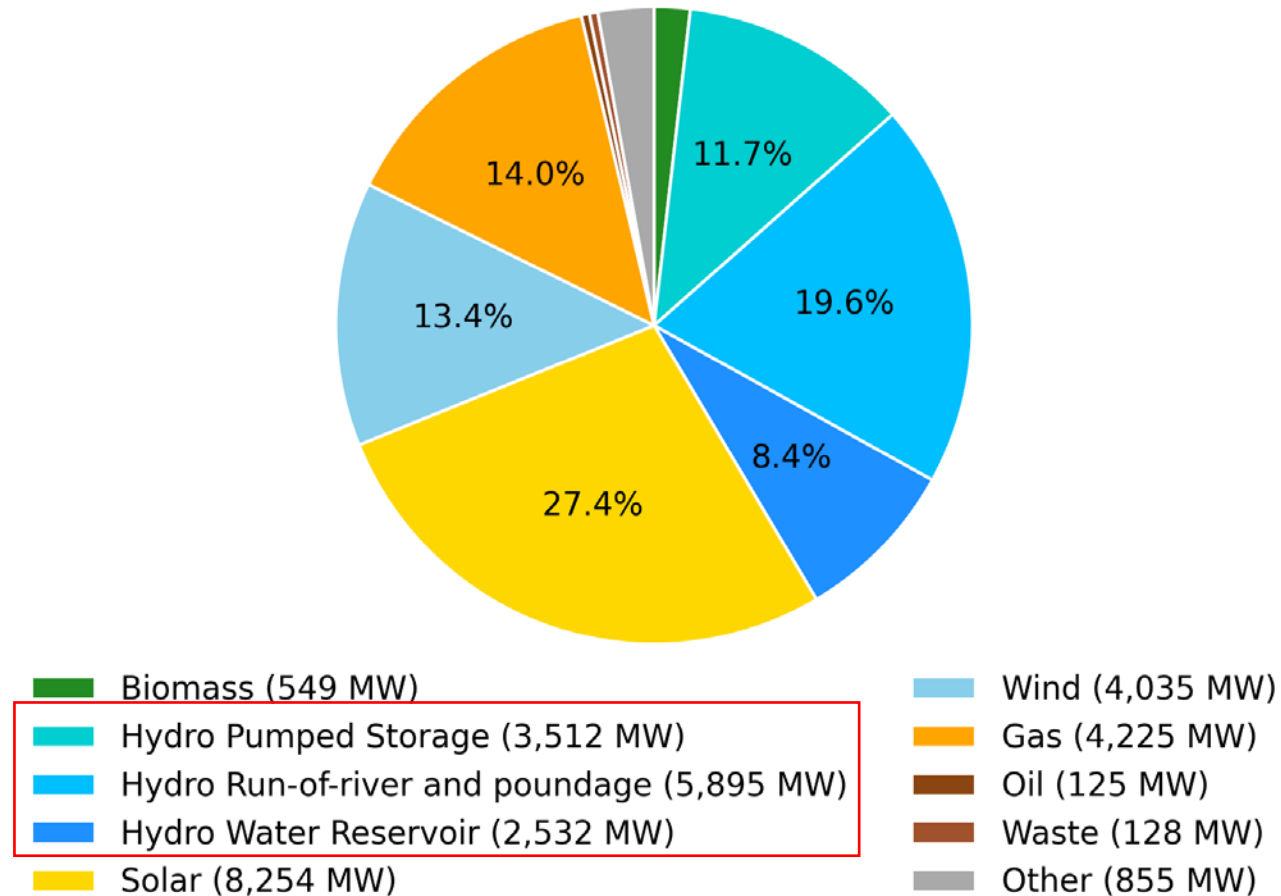


GDK Mellach 838 MW, Steiermark [4]



Generation

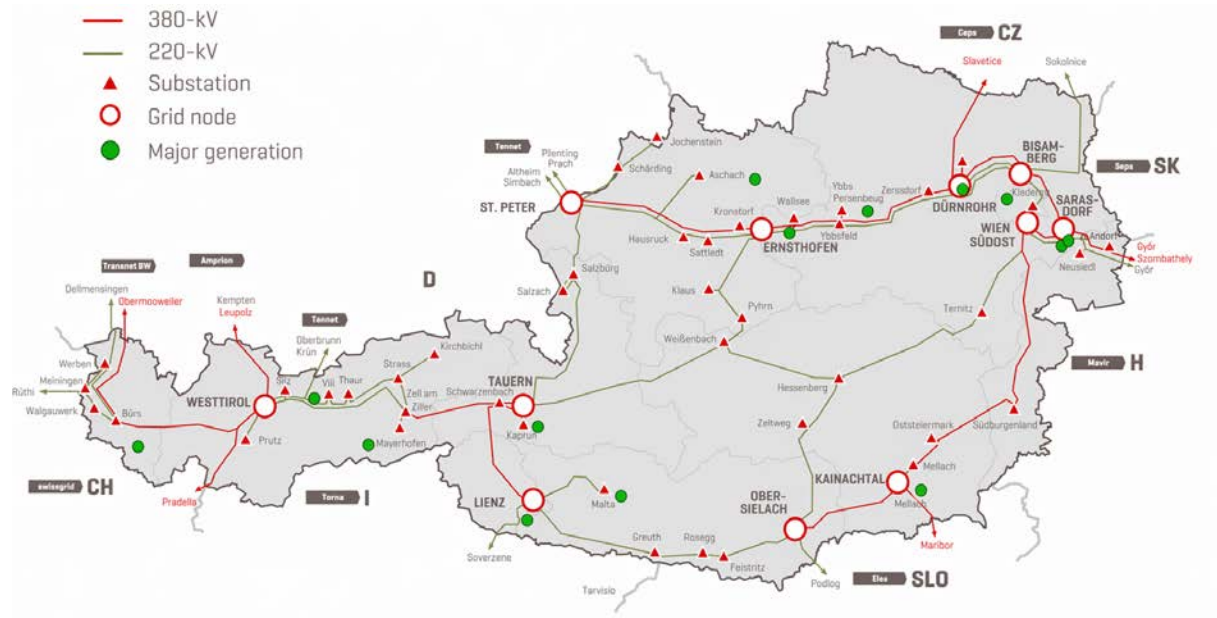
Installed capacity Austria (2025)



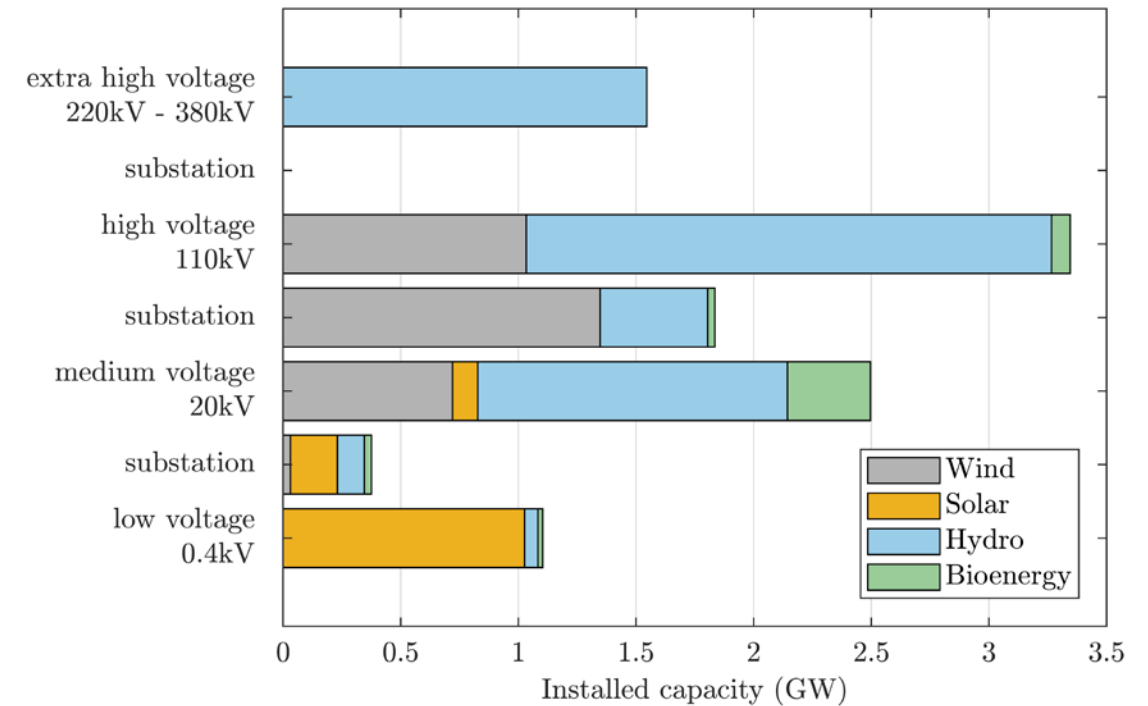
<https://markt.apg.at/en/transparency/generation/installed-power-plant-capacity/>

Where are all these generation being placed?

Renewable generation



Austria (2022)

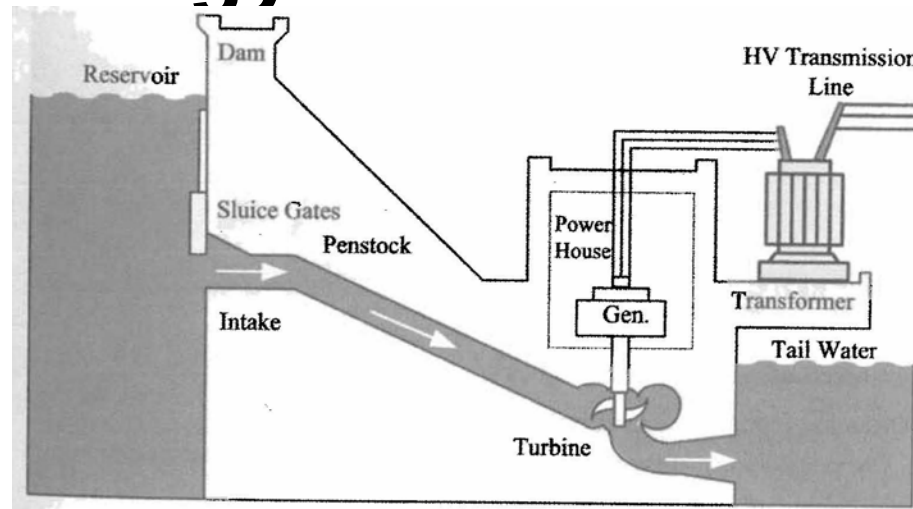


E-Control Austria. Flexibility study 2022. https://www.e-control.at/documents/1785851/1811582/20220207_Flexibilitaetsstudie_Bericht_FINAL.pdf/244c4f3c-c8a2-1114-c287-6d6b81d07817?t=1650436768857, 2022. Accessed: 2024-07-10.

Recommended task: Installed capacity of all generation?

Generation: technology

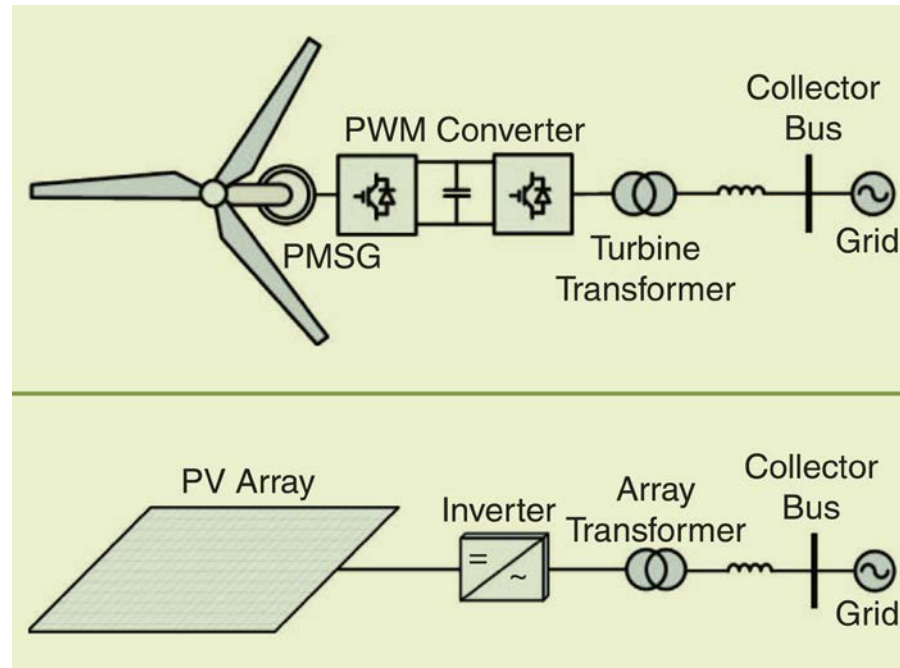
☐ Synchronous machine (e.g., hydro, gas)



(Saadat 2010)

- ☐ Controllable
- ☐ Predictable
- ☐ Natural inertia

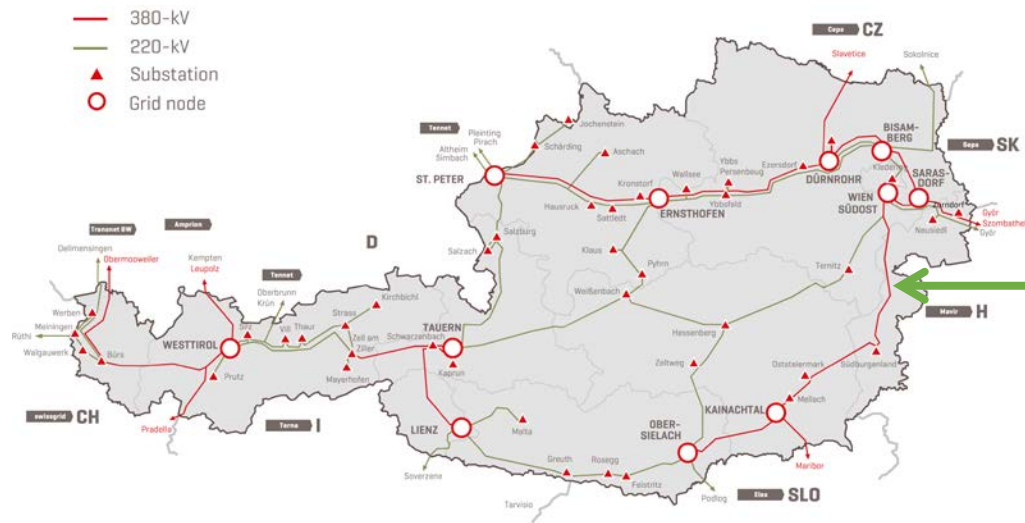
☐ Electronic-based (e.g., solar, wind)



- ☐ Variable
- ☐ Uncertain
- ☐ “Zero” inertia

(Kroposki et al. 2017)

Network: EHV, HV, MV, LV



Ground wire

Conductors (four)

Insulators

Tower

<https://ktn.lko.at/380-kv-stromtrasse-durch-k%C3%A4rnten-geplant+2400+4229556>

- ☐ In which **voltage level** are more **km** of lines?
- ☐ How much of the network is **visible**?
- ☐ How much is **AC** or **DC** network?

Öffentliches Netz (*) – Trassenlängen zum 31. Dezember 2024					
Spannungsebenen	Fernleitungen		Kabelleitungen		Summe
	km	Anteil in %	km	Anteil in %	km
380 kV	1.440	0,6	54	0,0	1.494
220 kV	1.847	0,7	8	0,0	1.856
110 kV	6.137	2,5	775	0,3	6.912
von 1 kV bis 110 kV	21.498	8,7	44.812	18,1	66.310
1 kV und darunter	25.707	10,4	145.291	58,7	170.998
insgesamt	56.630	22,9	190.941	77,1	247.571

(*) einschließlich Hoch- und Höchstspannungsleitungen von öffentlichen Erzeugern

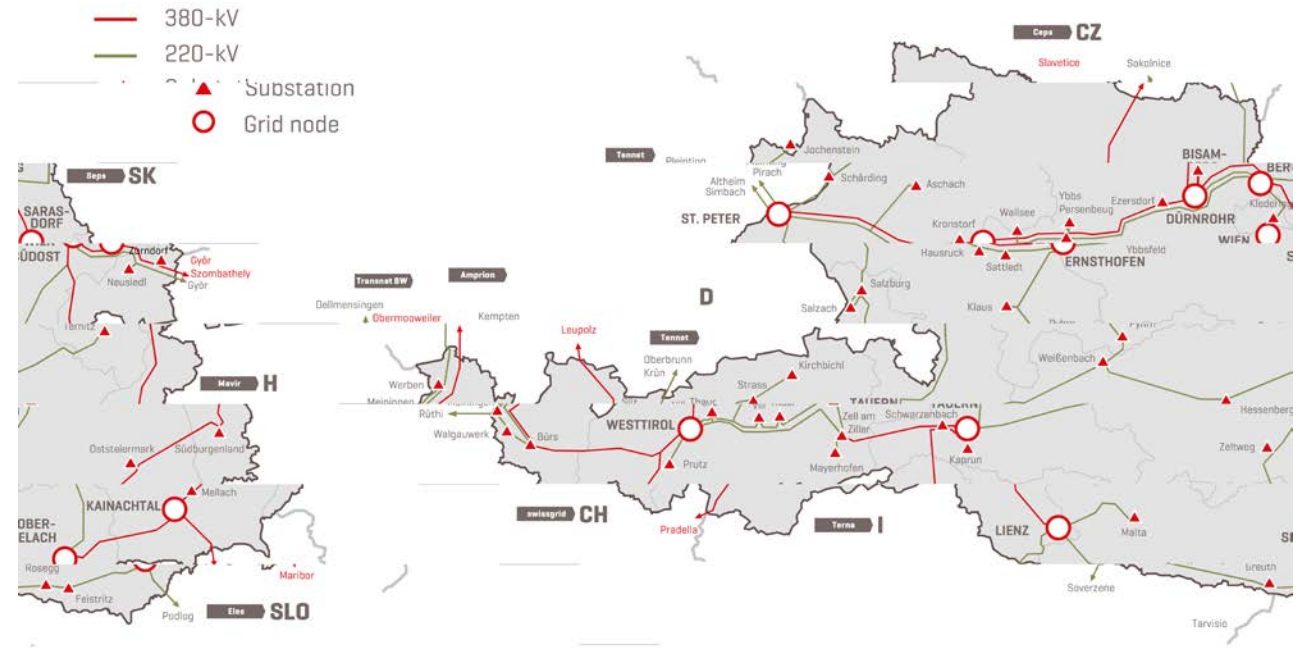
<https://www.e-control.at/documents/1785851/1811582/E-Control-Statistikbrochuere-2025-barrierefrei.pdf>

Transformers

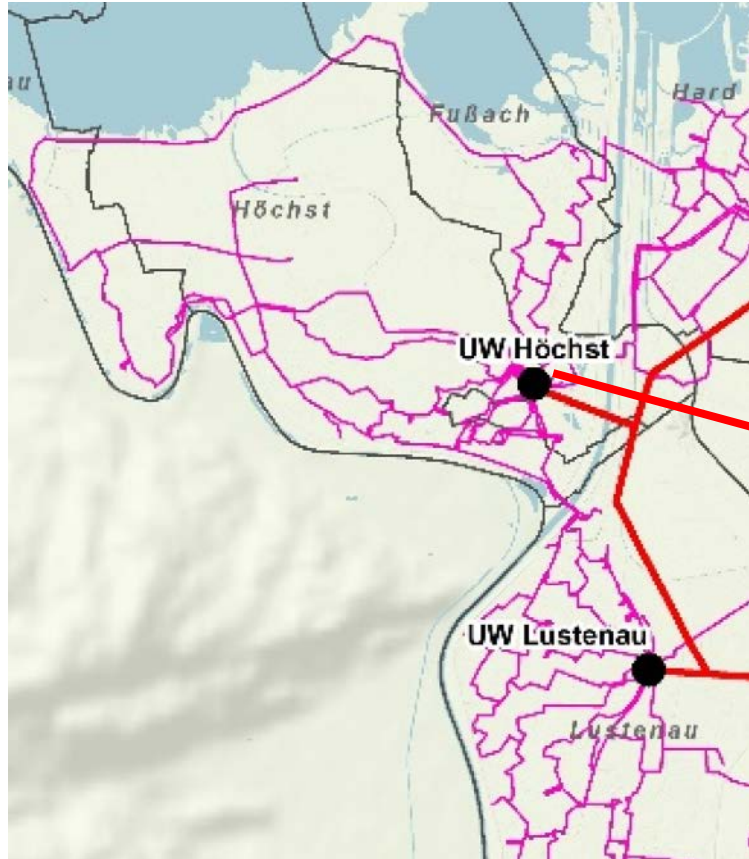


<https://www.vkw.at>

Medium-voltage (magenta)

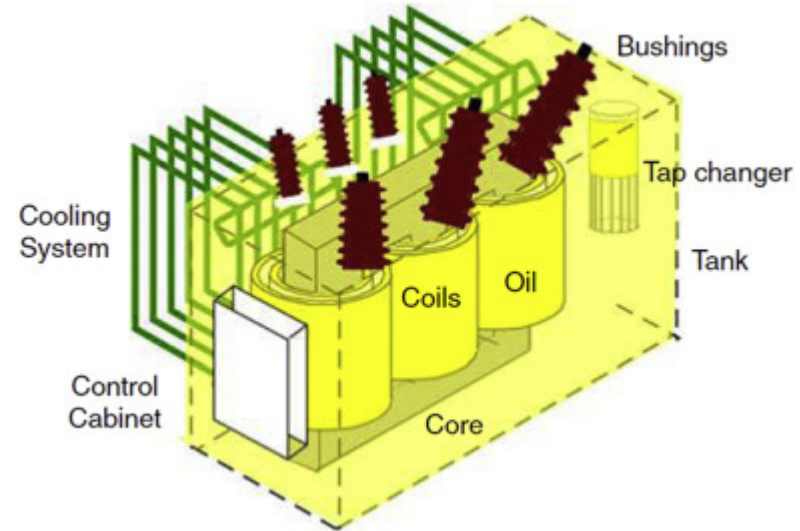


Transformers



<https://www.vkw.at>

Medium-voltage (magenta)

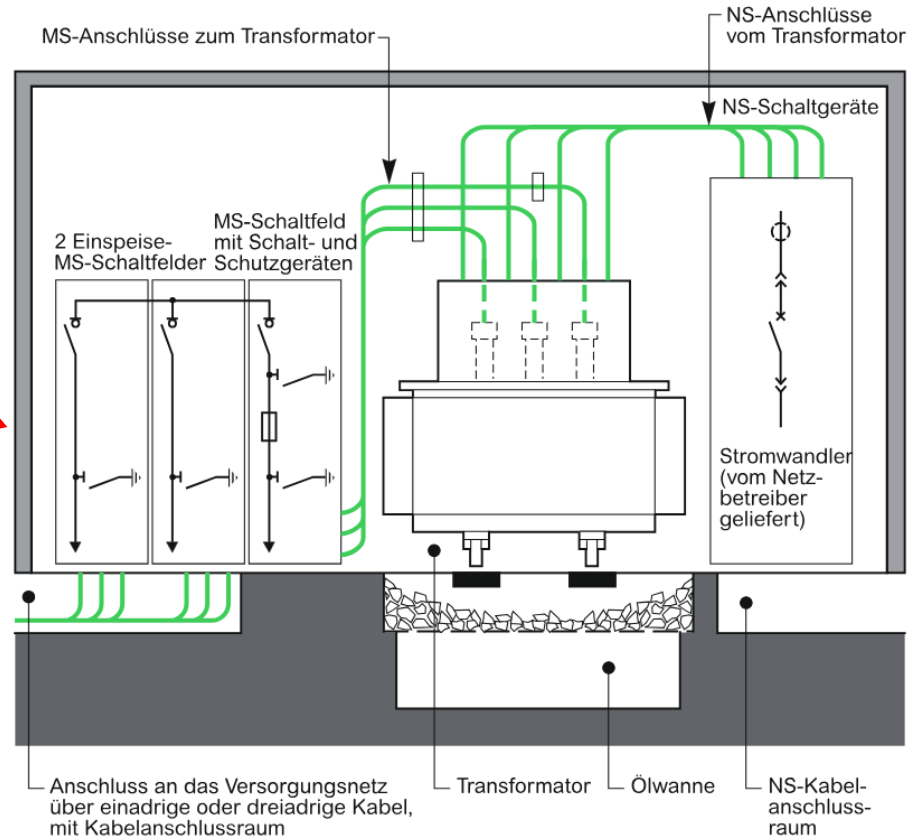
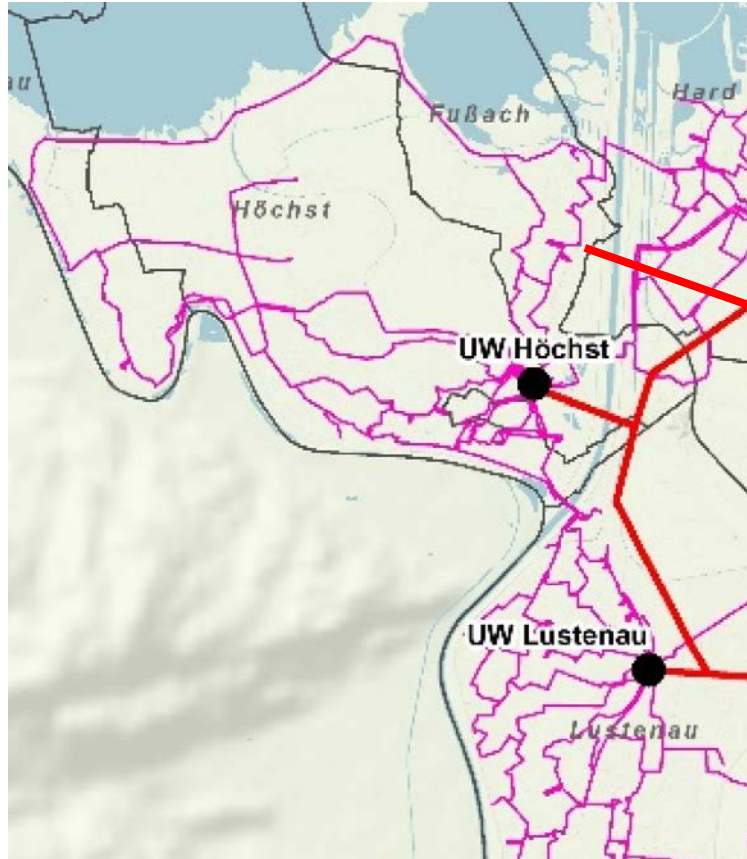


Murugan, R., & Ramasamy, R. <https://doi.org/10.1016/j.engfailanal.2018.10.011>



<https://www.google.com>

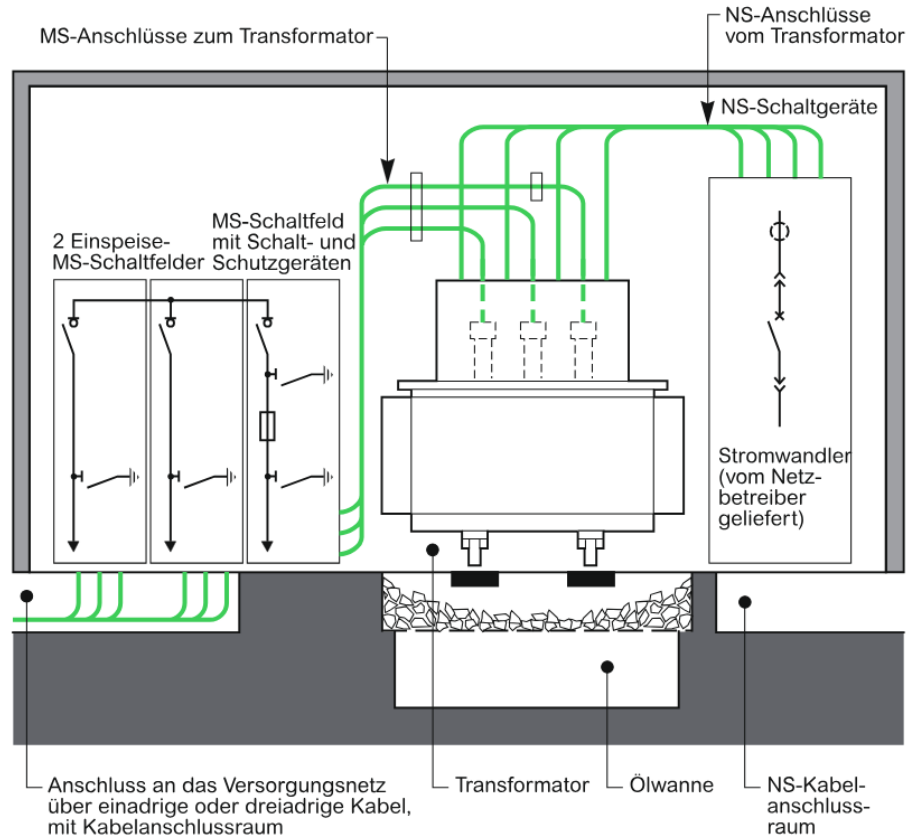
Transformers



<https://de.electrical-installation.org/dewiki/Innenraum-Netzstationen>

Medium-voltage (magenta)

Transformers



<https://de.electrical-installation.org/dewiki/Innenraum-Netzstationen>



Low-voltage (blue)

Transformers

How many transformers?

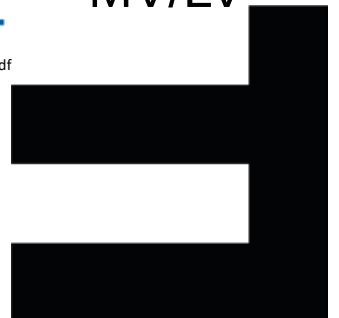
Öffentliches Netz – Umspann- und Schaltwerke zum 31. Dezember 2024		
Spannungsebenen	Anzahl der Transformatoren	Summenleistung in MVA
Oberspannung bis 220 kV	1.085	52.652
Oberspannung über 220 kV	97	34.655
Hochspannung zu Hoch-, Mittel- bzw. Niederspannung	1.182	87.307

EHV/HV
HV/MV

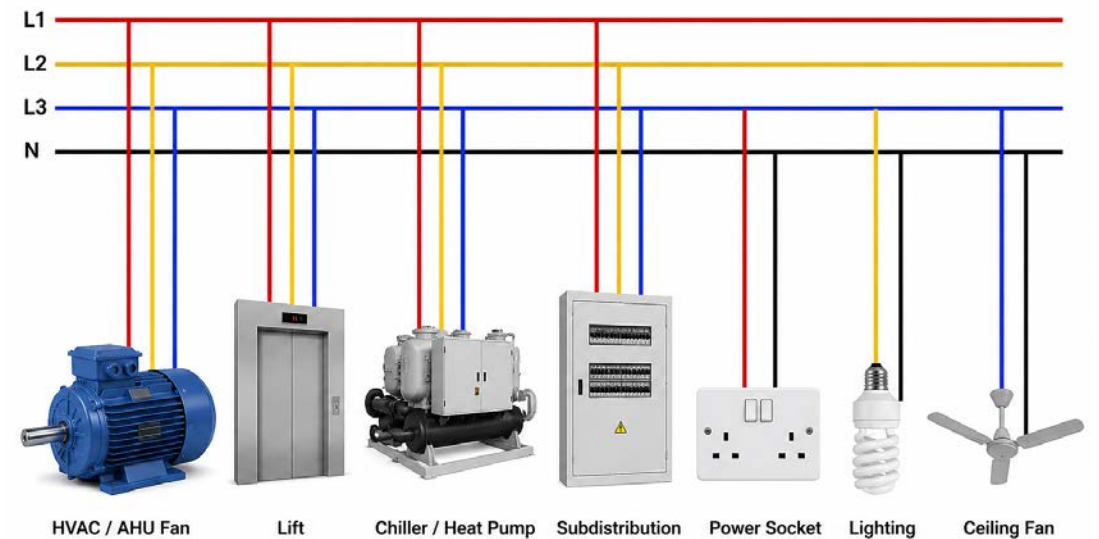
Öffentliches Netz – Transformatorstationen zum 31. Dezember 2024		
Spannungsebenen	Anzahl der Trafo-Stationen	Summenleistung in MVA
Mittelspannung zu Mittel- bzw. Niederspannung	84.384	38.147

MV/LV

<https://www.e-control.at/documents/1785851/1811582/E-Control-Statistikbroschuere-2025-barrierefrei.pdf>

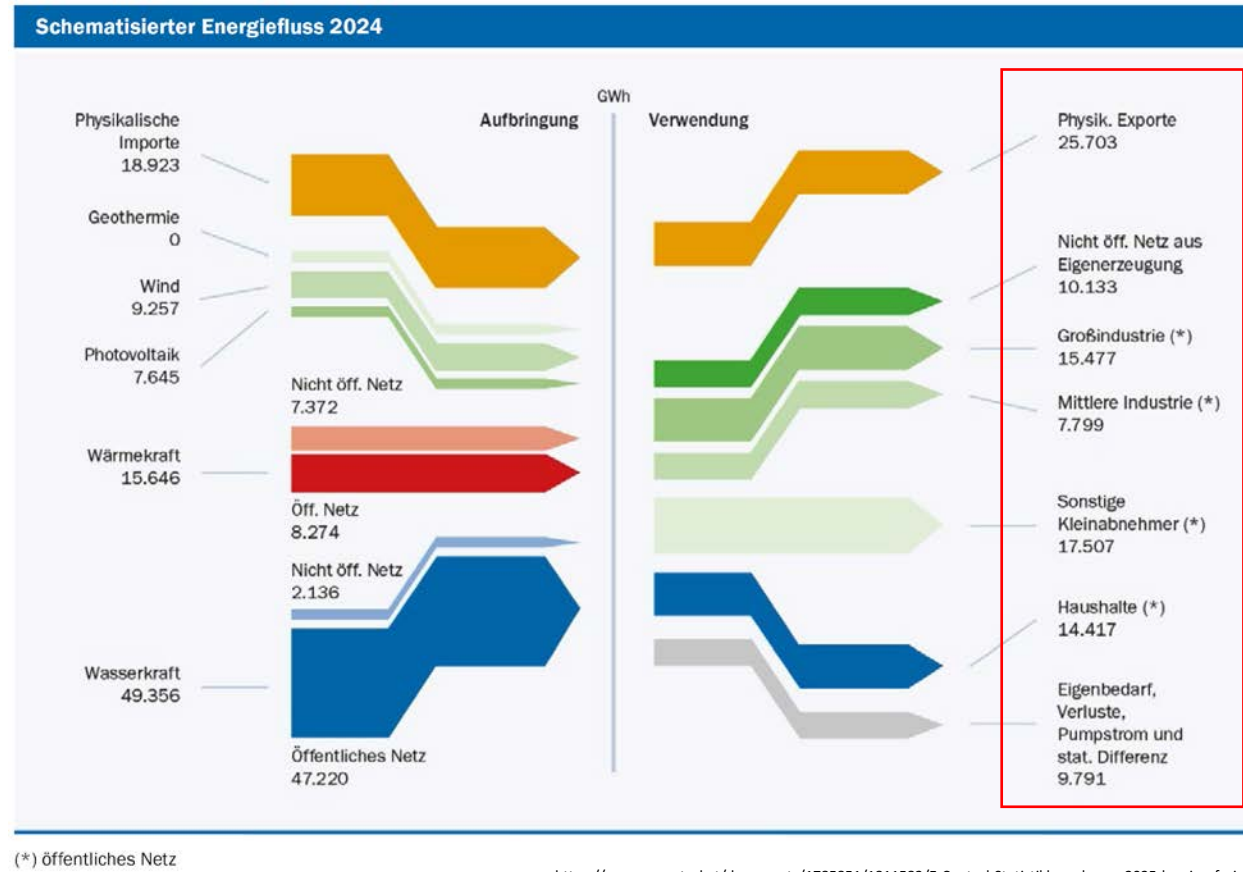


Demand



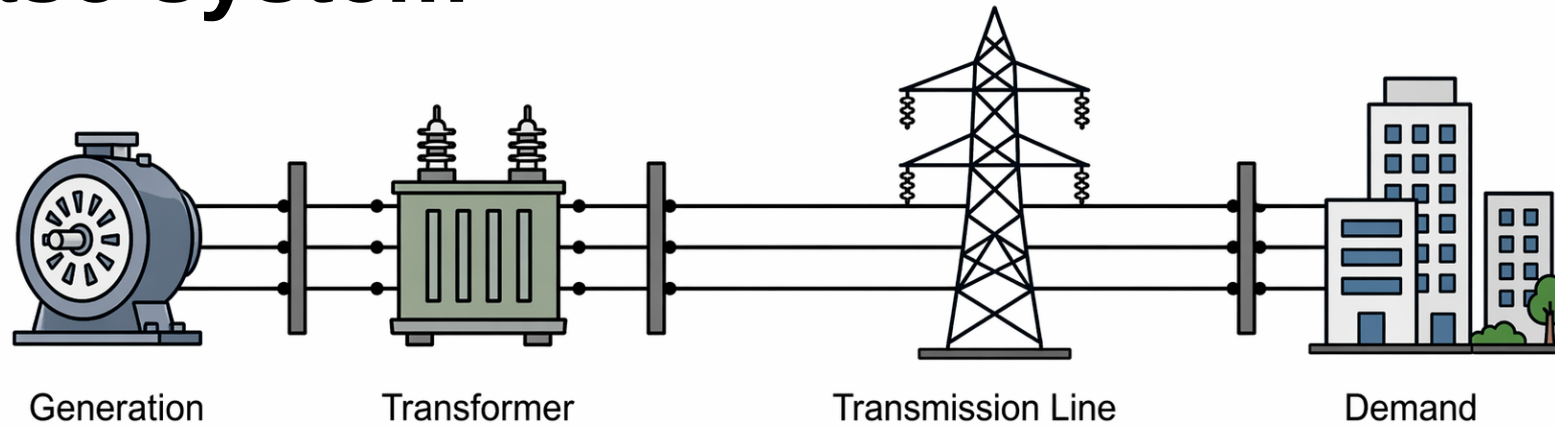
Demand

Energy (GWh)



<https://www.e-control.at/documents/1785851/1811582/E-Control-Statistikbrochuere-2025-barrierefrei.pdf>

Three-phase system

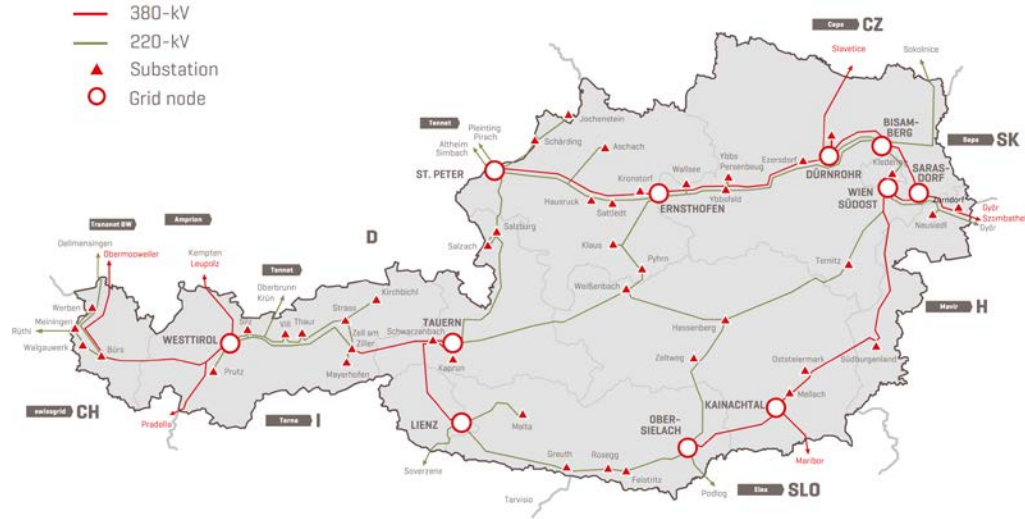


- ❑ **Balanced** three-phase system.
- ❑ **Steady state** condition (after transients or faults).

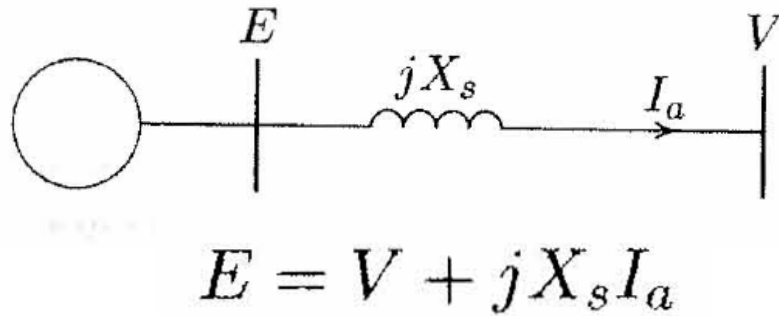
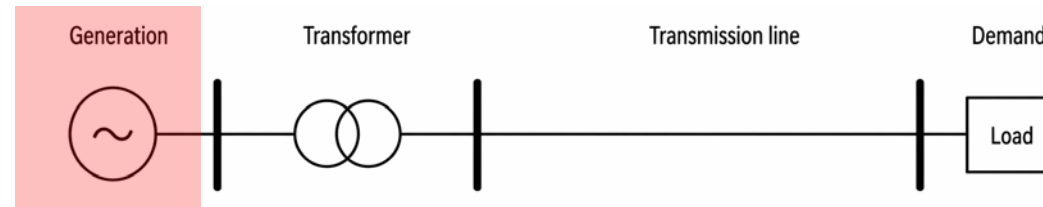
- Equivalent **single-phase** analysis
- Steady state models
- Generation = demands (frequency constant)



Three-phase system



Synchronous generation

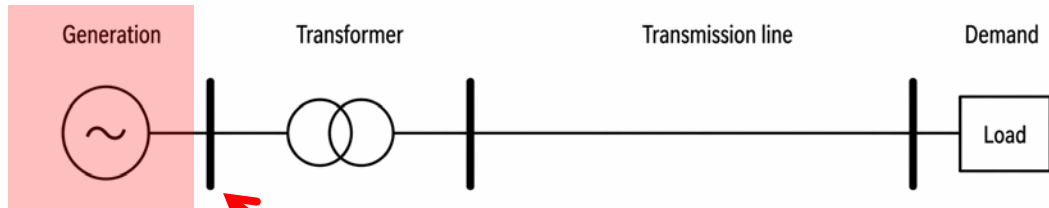


- V : terminal voltage
- E : internal (emf) voltage
- X_s : synchronous reactance
- I_a : armature current

Reactance Type	Study / Application	Timeframe
Synchronous (X_d)	Power Flow Analysis and steady-state stability ¹⁵ ¹⁶ . Used to determine field current and reactive power limits ¹⁶ ¹⁷ .	Steady-state (Sustained) ¹
Transient (X'_d)	Transient Stability Studies ¹⁴ . Used to analyze the machine's "swings" during the first second or two following a fault ¹⁸ ¹⁹ .	0.5 to 2.0 seconds ⁹ ²⁰
Subtransient (X''_d)	Short-Circuit Analysis and protection settings ²¹ . Used to compute the maximum initial current that circuit breakers must withstand ⁵ ²¹ .	First few cycles (0–0.1s) ⁹ ²²

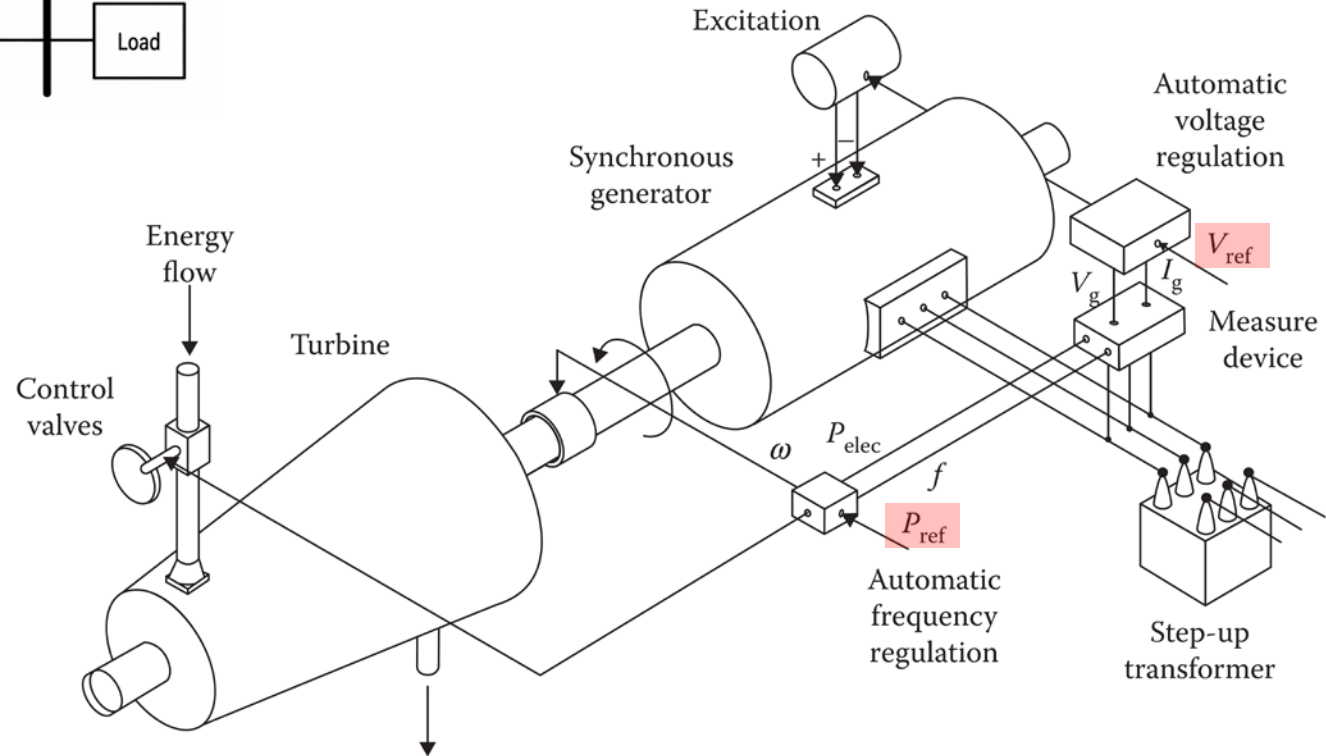
Saadat + Kundur + <https://notebooklm.google.com>

Synchronous generation



Fix voltage (V)
Fix active power (P)

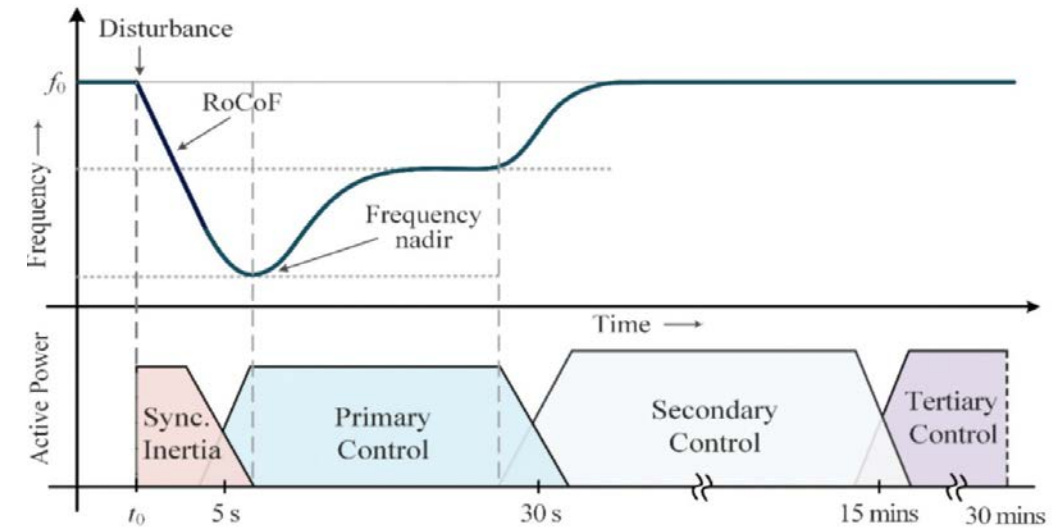
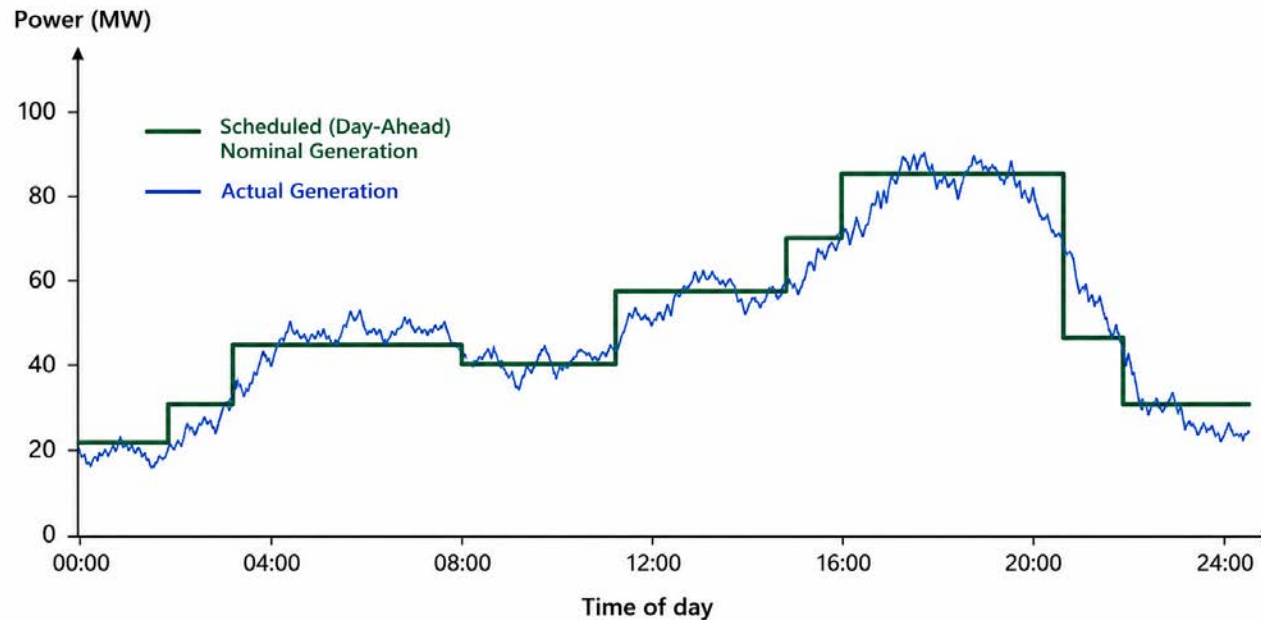
- ☐ Power control.
- ☐ Voltage control.



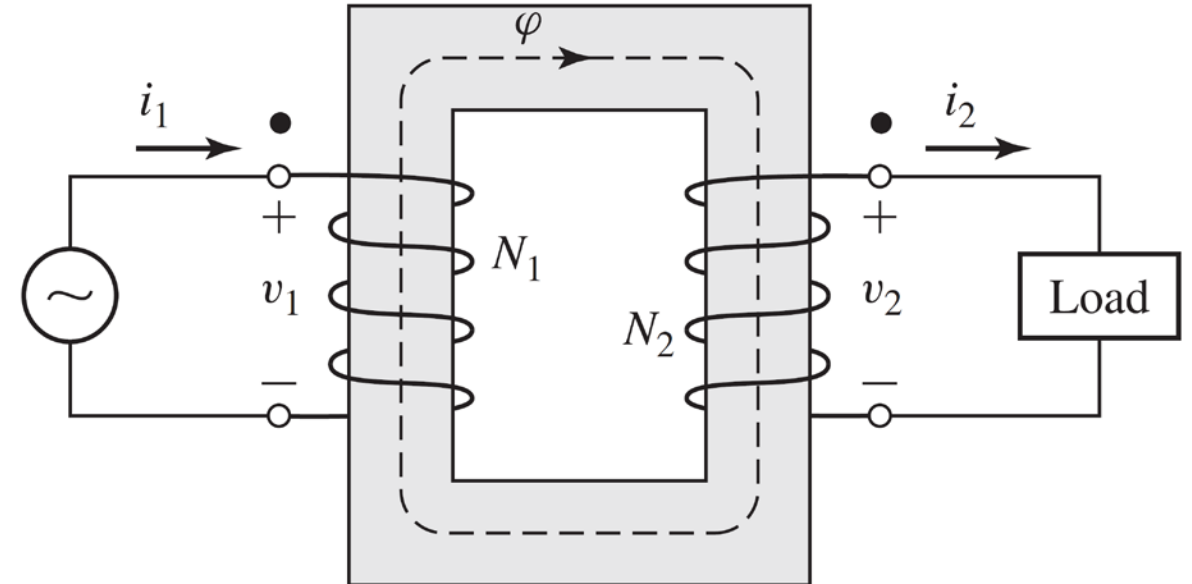
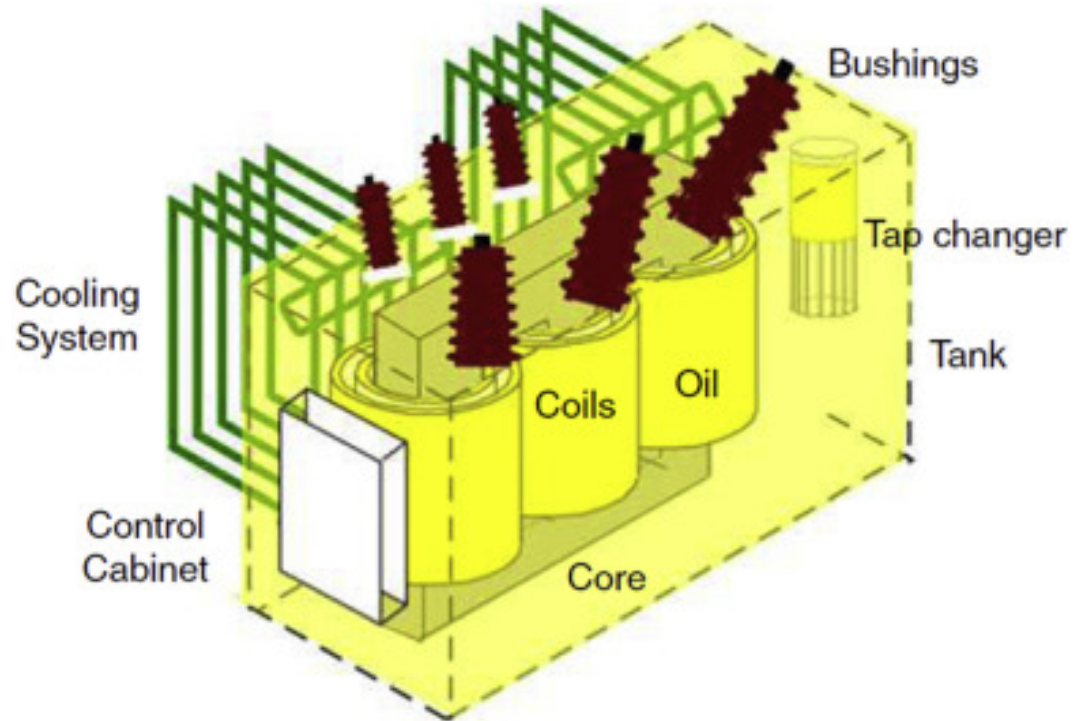
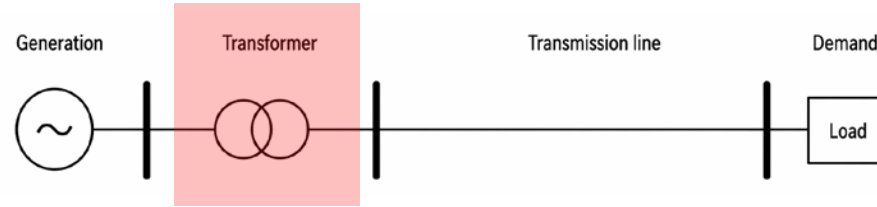
Book: Electric energy systems: analysis and operation, Gómez Expósito, Conejo and Canizares

Synchronous generation

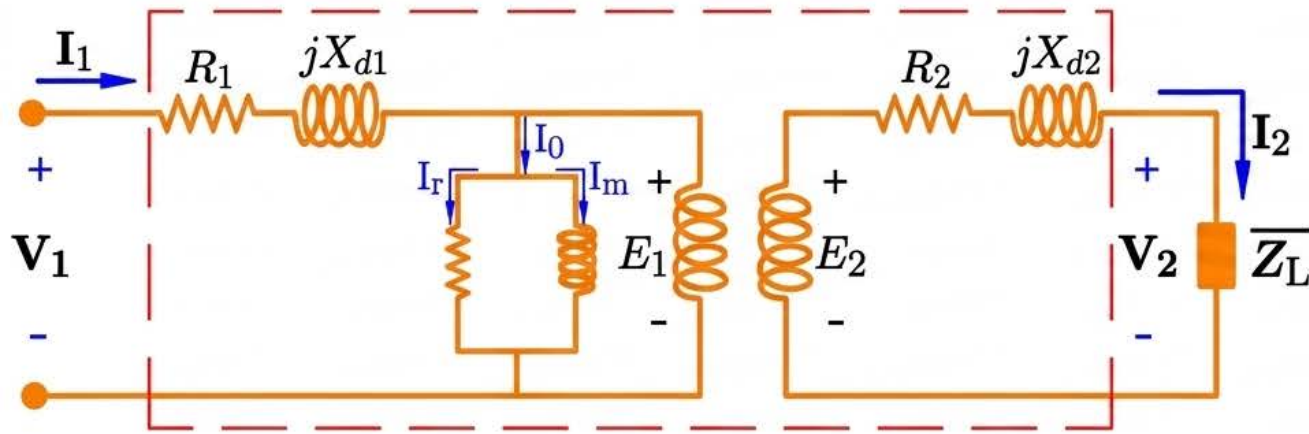
❑ Power control (scheduled)



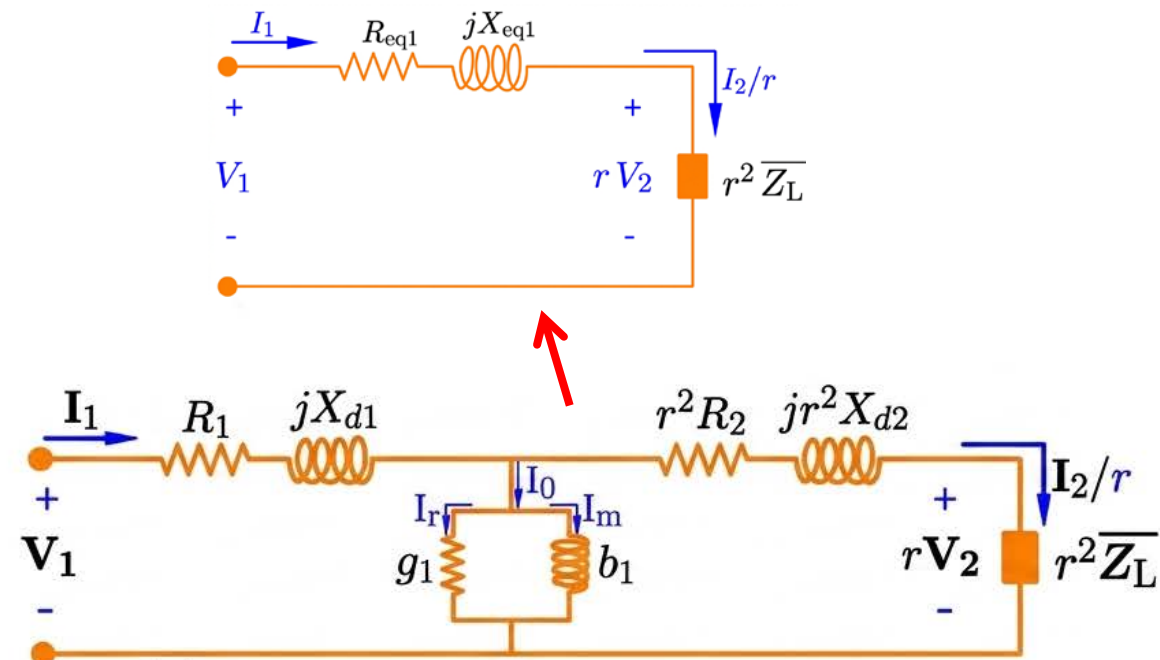
Transformer



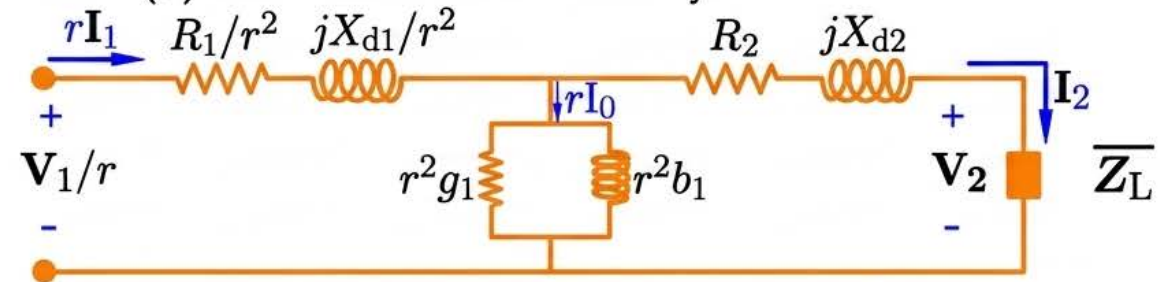
Transformer



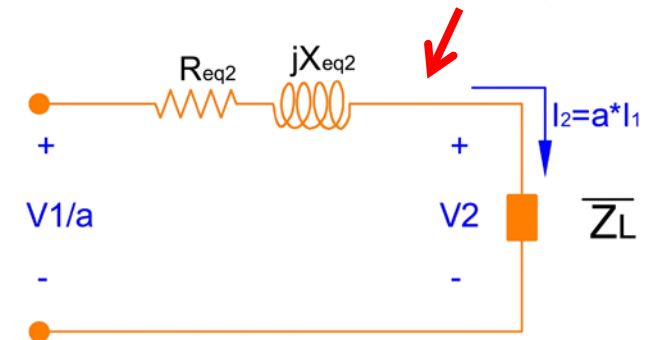
$$r = \frac{N_1}{N_2}$$



(a) Circuit Referred to Primary



(b) Circuit Referred to Secondary



Transformer parameters

1. SHORT CIRCUIT TEST

MEASUREMENTS

V_{sc} = Voltmeter reading

I_{sc} = Ammeter reading

P_{sc} = Wattmeter reading

RESULTS

Copper loss at I_{sc} ($\approx I_{rated}$)

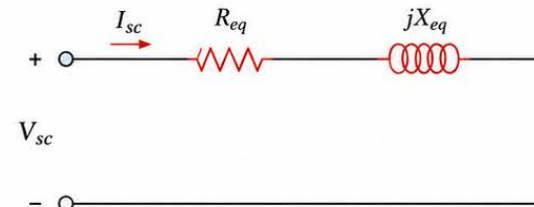
$P_{sc} \approx$ Copper loss

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}$$

$$R_{eq} = \frac{P_{sc}}{I_{sc}^2}$$

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

EQUIVALENT CIRCUIT (referred to primary)



$$R_{eq} = R_1 + R_2' \quad , \quad X_{eq} = X_{d1} + X_{d2}'$$

(All quantities referred to primary side)

- Secondary is short-circuited.
- Applied voltage is low (5–10% of rated).
- Iron loss is negligible (very small flux).
- The equivalent circuit reduces to series impedance only.

2. OPEN CIRCUIT (NO-LOAD) TEST

MEASUREMENTS

V_{oc} = Voltmeter reading

I_0 = Ammeter reading

P_{oc} = Wattmeter reading

RESULTS

Core (iron) loss

$P_{oc} \approx$ Core loss

$$\cos \phi_0 = \frac{P_{oc}}{V_{oc} I_0}$$

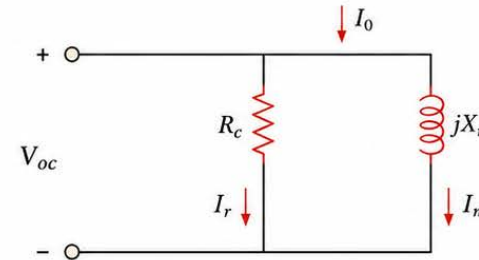
$$I_r = I_0 \cos \phi_0 \quad (\text{core-loss component})$$

$$I_m = I_0 \sin \phi_0 \quad (\text{magnetizing component})$$

$$R_c = \frac{V_{oc}}{I_r}$$

$$X_m = \frac{V_{oc}}{I_m}$$

EQUIVALENT CIRCUIT (referred to primary)



$R_c \rightarrow$ Core loss resistance
 $X_m \rightarrow$ Magnetizing reactance

- Secondary is open-circuited.
- Applied voltage is rated voltage.
- No-load current I_0 is small.
- Copper loss is negligible.
- The equivalent circuit reduces to shunt (excitation) branch only.

NOTES:

R_1, R_2 : Winding resistances (copper loss)

X_{d1}, X_{d2} : Leakage reactances

R_c : Core loss resistance

X_m : Magnetizing reactance

R_{eq} : Equivalent resistance

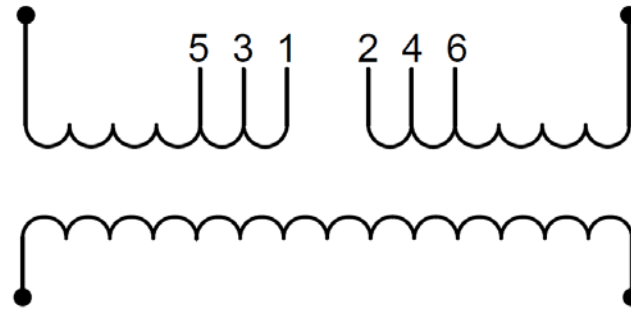
X_{eq} : Equivalent reactance

V : Voltage

I : Current

P : Power

Transformer Tap-Changer



Volts L-L	DETC Position	Leads Connected
144900	A	1 - 2
141450	B	2 - 3
138000	C	3 - 4
134550	D	4 - 5
131100	E	5 - 6

Figure 1-26- A typical winding schematic and connection diagram

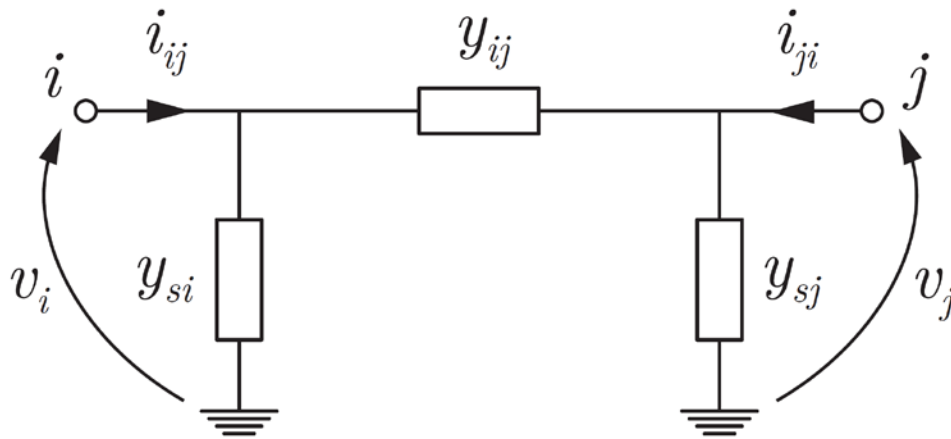
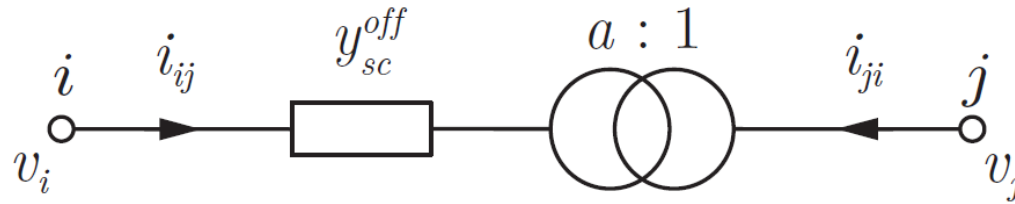
Shaarbafi, Karim. "Transformer modelling guide." Teshmont Consultants LP (2014).

How does it affect the model?

- Transformation ratio.
- The series impedance (and shunt impedance).



Transformer Tap-Changer



$$y_{ij} = -Y_{ij} = \frac{a(1+k)}{1+a^2k} y_{sc},$$

$$y_{si} = Y_{ii} + Y_{ij} = \frac{1-a+k(1-a)}{1+a^2k} y_{sc},$$

$$y_{sj} = Y_{jj} + Y_{ij} = \frac{a(a-1)(1+k)}{1+a^2k} y_{sc}.$$

<https://ieeexplore.ieee.org/document/8827633/>

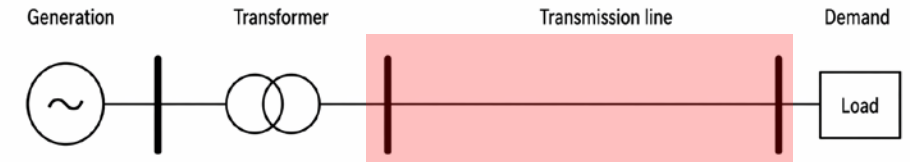
Transformer nameplate

CE		ABB	
Type TNOSCT-100/11PNS UNI			
No. 1LPL525913		Year of manufacture 2014	
Rated power 100 kV-A		No. of phases 3	
TRANSFORMER to specification EN (IEC) 60076-1			
Rated voltage [V]		Current [A]	Insulation level
HV	11000 $\pm 2 \times 2.5\%$	5.25	LI75 AC28
LV	415	139.10	AC3
Tap no.	HV voltage [V]	Connection symbol	Dyn11
1	11550	Cooling	ONAN
2	11275	Rated frequency	50 Hz
3	11000	Short-circuit imp.	3.88 %
4	10725	Load losses	1750 W
5	10450	No-load losses	145 W
Ambient temp. 40°C		Total mass	463 kg
Temperature rise of:		Mass of active part	279 kg
Windings	60K	Mass of oil	100 kg
Oil	55K	Type of oil	Mineral Nytro Taurus
		Oil to	IEC 60296
Windings material HV/LV: Al/Al		Core material: grain-oriented electrical steel	
Mass of windings: 53kg		Mass of core: 169kg	

<https://eepower.com/technical-articles/transformer-nameplate-details-and-sound-levels/>



Network: EHV, HV, MV, LV



Long Line Model

$(L) > 250$ km

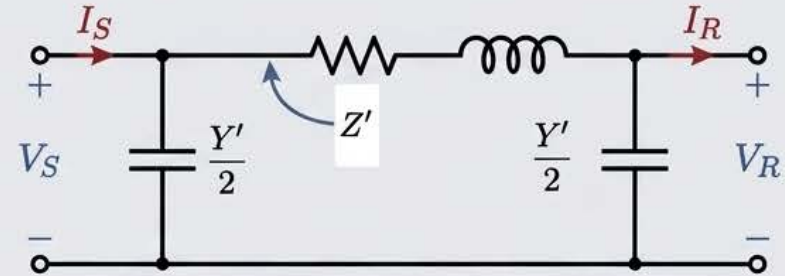
γ : propagation constant

ℓ : length

$$\gamma = \sqrt{ZY}$$

$$\frac{Y'}{2} = \frac{Y \tanh(\gamma\ell/2)}{\gamma\ell/2}$$

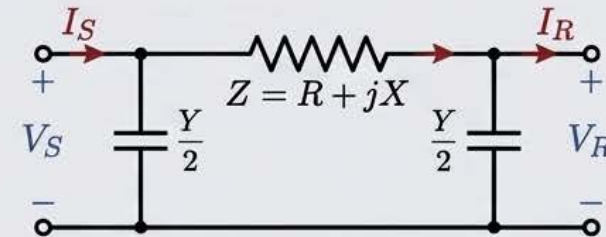
$$Z' = Z \frac{\sinh \gamma\ell}{\gamma\ell}$$



Medium Line Model

(L) : 80 – 250 km

$$Y = (g + j\omega C)\ell$$

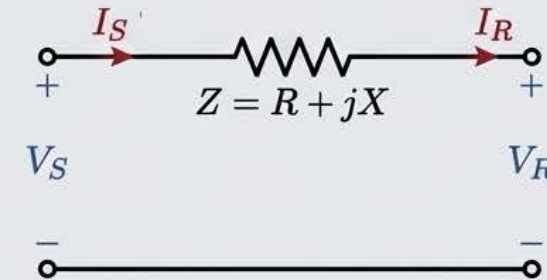


Short Line Model

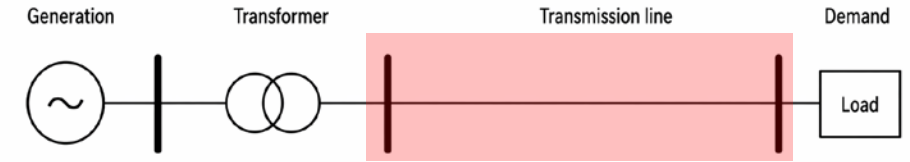
$(L) < 80$ km

Capacitance is neglected.

$$Z = (r + j\omega L)\ell$$



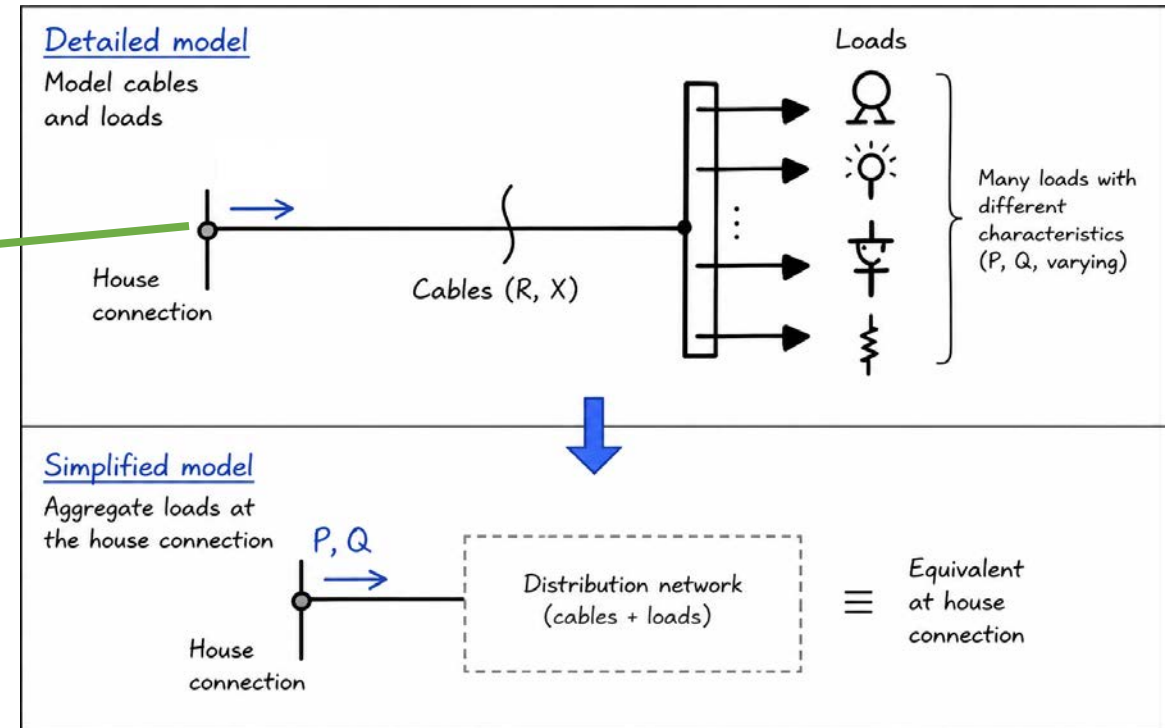
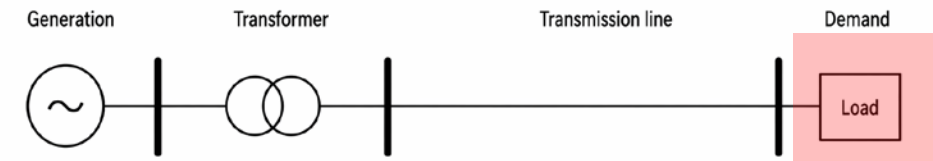
Network: EHV, HV, MV, LV



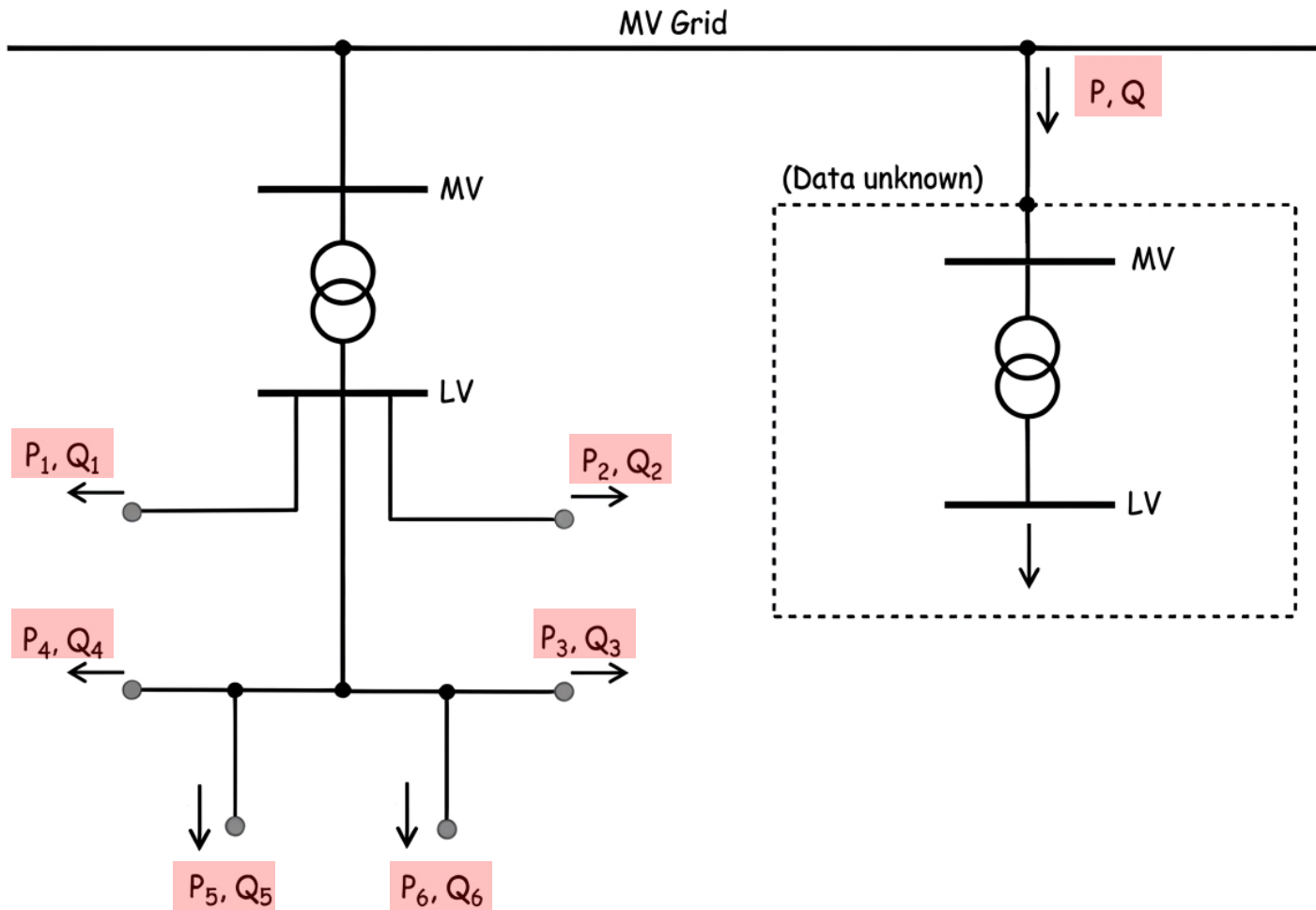
- Which model based on the voltage level (be careful: **rule-of-thumbs**):

	Overhead Lines	Underground Cables
Low Voltage	Short	Short
Medium Voltage	Short/Medium	Medium
High Voltage	Length-based	Medium/Long

Demand/load



Demand/load



Per-unit system

- Power system is **interconnected** with different **voltage levels**.
- Two base values are required to define a per-unit system:
- **S_b**: three-phase apparent power (VA)
- **V_b**: line-to-line voltage (V)

$$I_B = \frac{S_B}{\sqrt{3} V_B} \quad Z_B = \frac{(V_B)^2}{S_B} \quad Z_B = \frac{V_B / \sqrt{3}}{I_B}$$

- In a per-unit system, the different voltages levels disappear.

$$\bar{S}_{pu} = \frac{\bar{S}}{S_B} \quad V_{pu} = \frac{\bar{V}}{V_B} \quad \bar{I}_{pu} = \frac{\bar{I}}{I_B} \quad \text{and} \quad \bar{Z}_{pu} = \frac{\bar{Z}}{Z_B}$$

Per-unit system

- In a per-unit system, the factors **sqrt(3)** and **3** are eliminated.

$$S_{pu} = V_{pu} I_{pu}^*$$

$$V_{pu} = Z_{pu} I_{pu}$$

- Change of impedance from one system to other:

$$Z_{pu}^{new} = Z_{pu}^{old} \frac{S_B^{new}}{S_B^{old}} \left(\frac{V_B^{old}}{V_B^{new}} \right)^2$$

Transformer tests

Short-Circuit Test

$$Z_{eq} = R_{eq} + jX_{eq}$$

$$Z_b = \frac{V_n^2}{S_b}, \quad I_n = \frac{S_b}{\sqrt{3}V_n}$$

$$V_{sc} \frac{V_n}{\sqrt{3}} = Z_{eq} I_n$$

$$V_{sc}(p.u.) = \frac{Z_{eq}}{Z_b} = Z_{eq}(p.u.)$$

$$V_{sc}(\%) = Z_{eq}(p.u.) \times 100$$

$$P_{sc} = 3R_{eq}I_n^2$$

$$P_{sc}(p.u.) = \frac{3R_{eq}I_n^2}{S_b} = \frac{R_{eq}}{Z_b} = R_{eq}(p.u.)$$

$$P_{sc}(p.u.) = R_{eq}(p.u.)$$

Open-Circuit Test

$$Y_0 = G_{fe} + jB_m$$

$$Y_b = \frac{1}{Z_b}$$

$$I_0 = V_n Y_0$$

$$I_n = \frac{V_n}{\sqrt{3}Z_b}$$

$$I_0(p.u.) = \frac{I_0}{I_n} = \frac{Y_0}{Y_b} = Y_0(p.u.)$$

$$I_0(\%) = Y_0(p.u.) \times 100$$

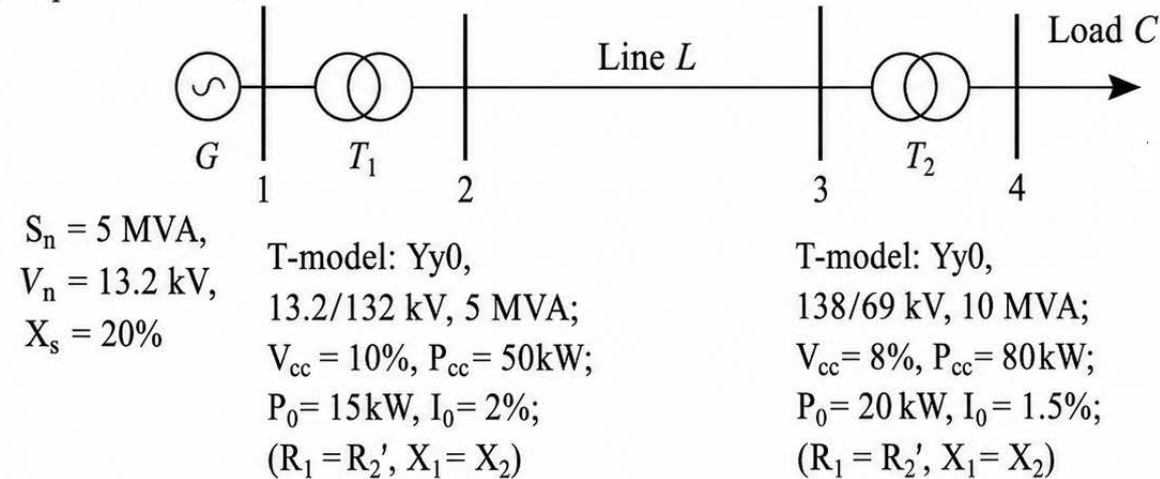
$$P_0 = V_n^2 G_{fe}$$

$$P_0(p.u.) = \frac{V_n^2 G_{fe}}{S_b} = \frac{G_{fe}}{1/Z_b} = G_{fe}(p.u.)$$

$$P_0(p.u.) = G_{fe}(p.u.)$$

Exercise

The figure below shows a single-line diagram of a power system consisting of a synchronous generator G , two transformers T_1 and T_2 , a transmission line, and a load C . The terminal voltage at bus 1 (generator terminals) is $V_1 = 13.2$ kV.



The transmission line L (connecting buses 2 and 3) has an impedance $Z_L = (10 + j100) \Omega$ and a total shunt admittance $Y_L = j3.0 \times 10^{-4}$ S. The load C (connected at bus 4) has an impedance $Z_C = 1100 \Omega$. Determine the current intensities throughout the network, voltage across the load, the power consumed by it.

References

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